Shahrzad Dehghani

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Solingen, NRW, Germany German Citizen
Portfolio LinkedIn ORCiD GitHub ResearchGate

Profile

Data Scientist & Data analyst with a strong research and industry background in machine learning. Skilled in statistical analysis, large-scale data analytics, and data visualization using Python, SQL, and BI tools. Developed, optimized and benchmarked deep learning and generative AI models for inverse design problems, transforming complex scientific and industrial datasets into accurate solutions. Most recently, led a self-driven analysis of 6.8M Amazon jewelry reviews, uncovering customer satisfaction drivers, brand performance trends, and pricing behavior through EDA, NLP, regression modeling, and interactive BI dashboards.

Skills

- **Programming Languages & Tools:** Python (Pandas, NumPy, SciPy, NLTK, etc.), SQL, Excel, MATLAB, C++
- **Machine Learning & Deep Learning:** PyTorch, Scikit-learn, TensorFlow, Keras, Natural Language Processing (NLP), Transformer Models, Hyperparameter Optimization (Optuna)
- **Data Analysis & Scientific Computing:** Data Wrangling, Exploratory Data Analysis (EDA), Statistical Analysis, Regression Analysis, KPI Development, A/B Testing Concepts, Large Datasets, Data-Driven Modeling of Physical Systems
- **Computer Vision & Imaging:** OpenCV, Hyperspectral Image Analysis, Image Processing
- **Data Visualization:** Power BI, Tableau, Matplotlib, Seaborn
- **Operating Systems & Development Environments:** Linux, Windows, macOS, Jupyter Notebook, JupyterLab, VS Code
- **High-Performance Computing (HPC):** Distributed CPU/GPU Computing, Job Scheduling (SLURM), Parallel Data Processing, Large-Scale Data Processing
- **Version Control:** Git, GitHub
- **Web Technologies:** HTML, CSS
- **Languages:** English (C1), German (B2 – Improving), Persian (Native)

Employment History

- **Machine Learning Researcher & Engineer (Researcher in Optoelectrical Engineering) · Bergische Universität Wuppertal, Germany** *08.2023 – Present*
  - Developed and benchmarked deep learning architectures, including a generative AI architecture, to solve an inverse design problem in nanophotonics: predicting structural parameters from reflection spectra with a mean absolute error (MAE) below 1 nm for both silver layer thickness and grating period. The models were trained on 15,000 simulated dataset and evaluated for robustness against measurement noise. The approach provides a data-driven alternative to costly physical characterization techniques. Implemented in Python/PyTorch with hyperparameter optimization (Optuna) and deployed on HPC environments using SLURM. Here are project explanation and paper.
  - Applied machine learning and data science techniques in a confidential industry collaboration with Louisenthal GmbH, including data preparation, predictive modeling, model optimization, and quantitative performance analysis.
  - Developed a Python based GUI application for the evaluation of university applicants; the software is currently in active use by the Bergische Universität Wuppertal.
  - Teaching & supervision: tutorial sessions in thin film technology, including explaining theoretical concepts and guiding practical exercises. Supervised bachelor thesis project and seminar presentations for undergraduate students.
- **Laboratory based research internship in experimental physics · Universität zu Köln** *11.2022 – 07.2023*
  - Conducted thin-film experiments under ultra-high vacuum (UHV) conditions and analyzed spectroscopy datasets (XPS, UPS, IPES, REELS) using Python and OriginLab.

Employment History (continued)

- Performed data preprocessing, cleaning, analysis, and visualization of complex experimental data to extract actionable scientific insights.
- Developed and optimized automation scripts for data processing, analysis workflows, and experimental evaluation, improving efficiency and reproducibility.

Research Assistant (Data Analysis & HPC — Master Thesis) · Forschungszentrum Jülich *9.2021 – 08.2022*

- Simulated, processed, and analyzed large-scale datasets from N-Body6++ star cluster models at the Jülich Supercomputing Centre using Python, C++, and HPC systems (SLURM). Managed the complete data pipeline, including data cleaning, processing, and visualization.
- Developed multi-scenario comparative statistical analyses across different model parameters to quantify environmental effects on protoplanetary disc sizes.
- Contributed to an independent research collaboration on the statistical analysis of observational data concerning the lifetime of protoplanetary disks; results were published in a peer-reviewed scientific publication (Paper).

Summer Internship · Universität Bonn, Germany *7.2020 – 10.2020*

- Analyzed hyperspectral image datasets using Python (SPy) to extract physical and chemical properties from multidimensional data.

Research Assistant · Institute for Research in Fundamental Sciences (IPM) in Tehran, Iran *02.2018 – 12.2019*

- Processed and analyzed astronomical observational datasets to create scientific stellar catalogs.
- Performed data cleaning, quality control, and validation of photometric measurements obtained from INT telescope observations.
- Applied statistical analysis techniques to identify variable stars and investigate the star formation history of galaxies.
- Integrated and cross matched multiple data sources, including telescope observations, Gaia catalogs, and external surveys, to improve data quality and completeness.
- Contributed to research projects that resulted in multiple scientific publications (Paper 1 und Paper 2).

Independent Projects

Mining insights from Amazon Jewelry Reviews *03.2026 – Present*

- From business analytics to insight generation using LLMs (Python, Power BI, SQL)
- **Goal**: Analyzing Amazon jewelry reviews to uncover customer satisfaction drivers, brand performance, and product issues using both data analytics and NLP/LLM techniques.
- **Part 1: Data Analysis & Business insights: brand benchmarking, price-segment analysis, and review-text mining.**
- Cleaned and wrangled a large scale dataset (6.8M reviews, 800k products, 23k brands) spanning 19 years of Amazon jewelry reviews, handling missing values, text normalization, and schema inconsistencies.
- Conducted EDA and NLP based review text analysis on Amazon jewelry dataset to uncover insights on product trends, brand performance, pricing, ratings, and customer sentiment.
- Identified key drivers of customer dissatisfaction by applying regression analysis. The analysis revealed that complaints related to durability and appearance had the strongest negative impact on product ratings across all price segments.
- Built an interactive Power BI dashboard visualizing brand KPIs, pricing trends, and category performance across 19 years of data.

Independent Projects (continued)

- **Part 2: Data Science (NLP & Transformer-based modeling)**
- Developed a transformer-based sentiment classification on review data. Fine-tuned DistilBERT and RoBERTa (Hugging Face) for 3-class sentiment classification; achieved 93% accuracy and macro F1 0.80 on held-out test data.
- Extending the project with topic modeling, aspect-based sentiment analysis, and fake review detection.
- The full analysis, codes, results, visualizations, and thorough explanations are available in my Github or my Portfolio.

Education

M.Sc. in Physics, Universität zu Köln — *04.2020 – 03.2023*
Thesis title: *Role of gas expulsion time on cluster expansion. [Jülich Supercomputing Center (JSC)]*

M.Sc. in Theoretical Physics, Alzahra University, Tehran, Iran — *10.2014 – 07.2016*
Full scholarship as a top student in the national entrance exam for master's programs

B.Sc. in Physics, Shahid Beheshti University, Tehran, Iran — *10.2009 – 08.2014*
Specialization : Atomic and Molecular Physics - Full scholarship as a top 1% student in the national Iranian university entrance exam.

Research Publications

1. S. Dehghani, C. Knoth, et al., "Data-driven inverse design of hybrid waveguide gratings using reflection spectra via tandem networks and conditional vaes," *MDPI - Optics*, Nov. 2025. DOI: 10.3390/opt6040061.
2. T. Parto, S. Dehghani, et al., "The isaac newton telescope monitoring survey of local group dwarf galaxies. v. the star formation history of sagittarius dwarf irregular galaxy derived from long-period variable stars," *The Astrophysical Journal*, Jan. 2023. DOI: 10.3847/1538-4357/aca471.
3. S. Pfalzner, S. Dehghani, et al., "Most planets might have more than 5 myr of time to form," *The Astrophysical Journal Letters*, Oct. 2022. DOI: 10.3847/2041-8213/ac9839.
4. T. Parto, S. Dehghani, et al., "Int monitoring survey of local group dwarf galaxies: Star formation history and chemical enrichment," *Communications of the Byurakan Astrophysical Observatory*, 2020, ISSN: 2579-2776. DOI: 10.52526/25792776-2020.67.2-232.

Volunteering

Build Your Future — *06.2017 - 08.2017*
Held Soft Skills Workshops (AIESEC in Baku, Azerbaijan)